



Mazen Elnaghy

AI ENGINEER • DATA SCIENTIST • DATA ENGINEER

Port Said, Egypt | [linkedin.com/in/mazen-elnaghy](https://www.linkedin.com/in/mazen-elnaghy) | github.com/mazenelnaghy-code | mazen.elnaghy@gmail.com

LINKEDIN HEADLINE

AI / Machine Learning Engineer | Python, C++ | Data Engineering Trainee @ DEPI | Data Science & UI/UX

ABOUT

I'm an AI / Machine Learning Engineer who enjoys building intelligent systems that actually work outside the lab.

I ranked first in my department in AI and Machine Learning Engineering for the 2024–2025 academic year, and I hold a 3.95/4.0 GPA.

I have a broad foundation across the AI stack — classical machine learning, deep learning, computer vision, natural language processing, and large language models — which lets me choose the right approach for a problem rather than forcing one technique onto everything. I work mainly in Python and C++, and I also design clean, user-friendly interfaces so the systems I build are practical for the people who actually use them.

Most of my hands-on work has been in computer vision on edge devices. I've trained YOLO models for real-time object detection and deployed them on hardware such as the Raspberry Pi, including a dog detection and deterrent system built to protect kindergartens and residential areas from stray dog encounters, which was recognized as an outstanding engineering project at the EAST Engineering Expo.

My major gave me a solid foundation in data engineering, and I have working knowledge of data science alongside it — from cleaning and modeling data to drawing insights from it. I'm currently building on that in the Data Engineering track at the Digital Egypt Pioneers Initiative (DEPI), where I work with SQL, ETL pipelines, and cloud data tools.

What I enjoy most is the messy middle of a project: the point where a model works on clean sample data but not on real inputs. Solving that gap — better data, better preprocessing, smarter design — is where the real engineering happens, and it's the part I keep coming back to.

Beyond my degree, I've pursued additional training through the National Telecommunication Institute and Huawei's technical programs, and I've competed in national initiatives such as Digitopia. I work comfortably in both Arabic and English in professional settings.

I'm looking for opportunities where I can build and deploy machine learning models in real production environments.

Feel free to reach out — I'd be glad to connect.

EXPERIENCE

Programming Intern — Suez Canal Authority

August 2026 · Port Said, Egypt

- Completed a one-month intensive software development training program within a live engineering environment.
- Developed and debugged applications in Python and C++, applying object-oriented design principles and structured problem-solving.
- Designed UI/UX layouts and user flows to make technical tools clear and easy to operate for non-technical staff.
- Collaborated with engineers and technical staff to translate real operational requirements into working software.
- Practiced professional documentation and code readability standards used in an organizational setting.

EDUCATION

East Port Said National University

B.Sc. in Artificial Intelligence Engineering · 2023 – 2028 (Expected)

GPA: 3.95 / 4.0 · Ranked 1st in the AI & Machine Learning Engineering department (2024–2025)

Relevant coursework: Machine Learning · Deep Learning · Computer Vision · Data Structures & Algorithms · Database Systems · Object-Oriented Programming · Computer Architecture · Control Systems Engineering · Signals & Systems · Robotics · Linear Algebra & Probability

Scholarships

Data Engineering Track — Digital Egypt Pioneers Initiative (DEPI), MCIT

June 2026 – February 2027 · Egypt

- Completing an intensive professional track covering data engineering fundamentals, SQL, ETL pipelines, data warehousing, and cloud data platforms.
- Building hands-on projects that transform raw, unstructured data into clean, analysis-ready datasets.
- Applying data modeling and normalization techniques to design efficient, scalable database schemas.
- Completing parallel professional-skills training in business communication, emotional intelligence, and customer excellence.

PROJECTS

AI Dog Detection & Deterrent System

Built a computer vision system that detects dogs in a live camera feed and automatically emits a deterrent sound to keep them away from the area. Designed for safety applications such as protecting kindergartens, schoolyards, and residential entrances from stray dog encounters.

Trained a YOLO object detection model and deployed it on a Raspberry Pi for real-time inference on the edge, integrating the detection output with an audio trigger and tuning the system to minimize false alarms while running within the device's limited compute budget. Recognized as an outstanding engineering project at the EAST Engineering Expo, Faculty of Engineering.

Skills: Python · Computer Vision · Object Detection · YOLO · Raspberry Pi · Edge Computing

Real-Time Orange Ball Detection System

Trained a custom YOLO model to detect and track an orange ball in a live camera feed. Handled the complete pipeline end to end: dataset collection and annotation, model training and evaluation, and real-time inference in Python with OpenCV. Tuned the model to maintain detection accuracy under varying lighting and background conditions.

Skills: Python · YOLO · Ultralytics · OpenCV · Dataset Annotation

Egypt Air Quality Data Warehouse

github.com/mazenelnaghy-code/egypt-air-quality · Live dashboard: mazenelnaghy-code.github.io/egypt-air-quality

Built an automated ingestion and transformation pipeline in Python that collects hourly air quality and weather data for 8 Egyptian cities from the Open-Meteo API, and stores it as a star schema in DuckDB.

Orchestrated the pipeline on a 6-hourly schedule with GitHub Actions, publishing a live dashboard and an open CSV dataset with no server and no hosting cost. Implemented 12 data quality assertions that run on every build and block publication whenever a check fails, and made rebuilds idempotent by de-duplicating overlapping API windows.

Skills: Python · SQL · DuckDB · ETL & ELT Pipelines · Dimensional Modeling · Data Warehousing · Data Quality Validation · GitHub Actions

Prison Management Database System

github.com/mazenelnaghy-code/prison-management-system

Designed and implemented a relational database to manage prison records, including inmates, cells, staff assignments, and visit logs. Built the entity-relationship model from requirements, normalized the schema to eliminate redundancy, and wrote SQL queries for record retrieval, filtering, and reporting. Built a role-based web application over the same data model (Node.js, Express, REST API), with Admin, Guard, and Staff permissions enforced server-side on every endpoint.

Skills: SQL · Database Design · ER Modeling · Data Normalization · Query Optimization · Node.js · Express · REST API

CERTIFICATIONS

- Data Engineering Track — Digital Egypt Pioneers Initiative (DEPI), MCIT · June 2026 – February 2027 (in progress)
- Certificate of Excellence, 1st Rank in AI & Machine Learning Engineering — East Port Said National University · 2025
- Certificate of Recognition, Outstanding Engineering Project (EAST Engineering Expo) — East Port Said National University · 2026
- Quality Assurance in Education Workshop — East Port Said National University · June 2025

SKILLS

Programming: Python · C++ · SQL · MicroPython

AI / ML: Machine Learning · Deep Learning · Neural Networks · TensorFlow · Scikit-learn · NumPy · pandas · Natural Language Processing · Large Language Models

Computer Vision: YOLO · OpenCV · Object Detection · Image Processing · Real-Time Inference

Data: Data Science · ETL & ELT Pipelines · Data Modeling · Data Analysis · DuckDB · Data Warehousing · Dimensional Modeling (Star Schema) · Data Quality & Validation

Robotics & Edge: Raspberry Pi · Edge Computing · ROS 2

Design: UI/UX Design · Wireframing · User-Centered Design

Tools: Git · GitHub Actions (CI/CD) · Jupyter Notebook · PyCharm · Anaconda · VS Code

Soft Skills: Problem Solving · Analytical Thinking · Technical Communication · Presentation Skills · Teamwork & Collaboration · Adaptability · Time Management · Attention to Detail · Self-Learning · Emotional Intelligence · Leadership

COURSES

- Python Programming Course — Codezilla (paid online course)
- Machine Learning Program — National Telecommunication Institute (NTI)
- Data Engineering Track — Digital Egypt Pioneers Initiative (DEPI), MCIT
- Huawei Technical Training Programs — Artificial Intelligence & ICT tracks

HONORS & ACTIVITIES

- 1st Rank — AI & Machine Learning Engineering department, East Port Said National University (2024–2025)
- Outstanding Engineering Project — EAST Engineering Expo, Faculty of Engineering (2026)
- Digitopia 2025 — Participant, national digital innovation competition

LANGUAGES

- Arabic — Native or bilingual proficiency
- English — Professional working proficiency (C1)